

AI SHOOTER CHALLENGE

Java Programming Challenge

- ⑧ Object-Oriented Design
- ⑧ BFS Pathfinding AI
- ⑧ Real-time Game
- ⑧ Java Swing (VS Code)

BUILD SMART. CODE STRONG. WIN THE CHALLENGE!

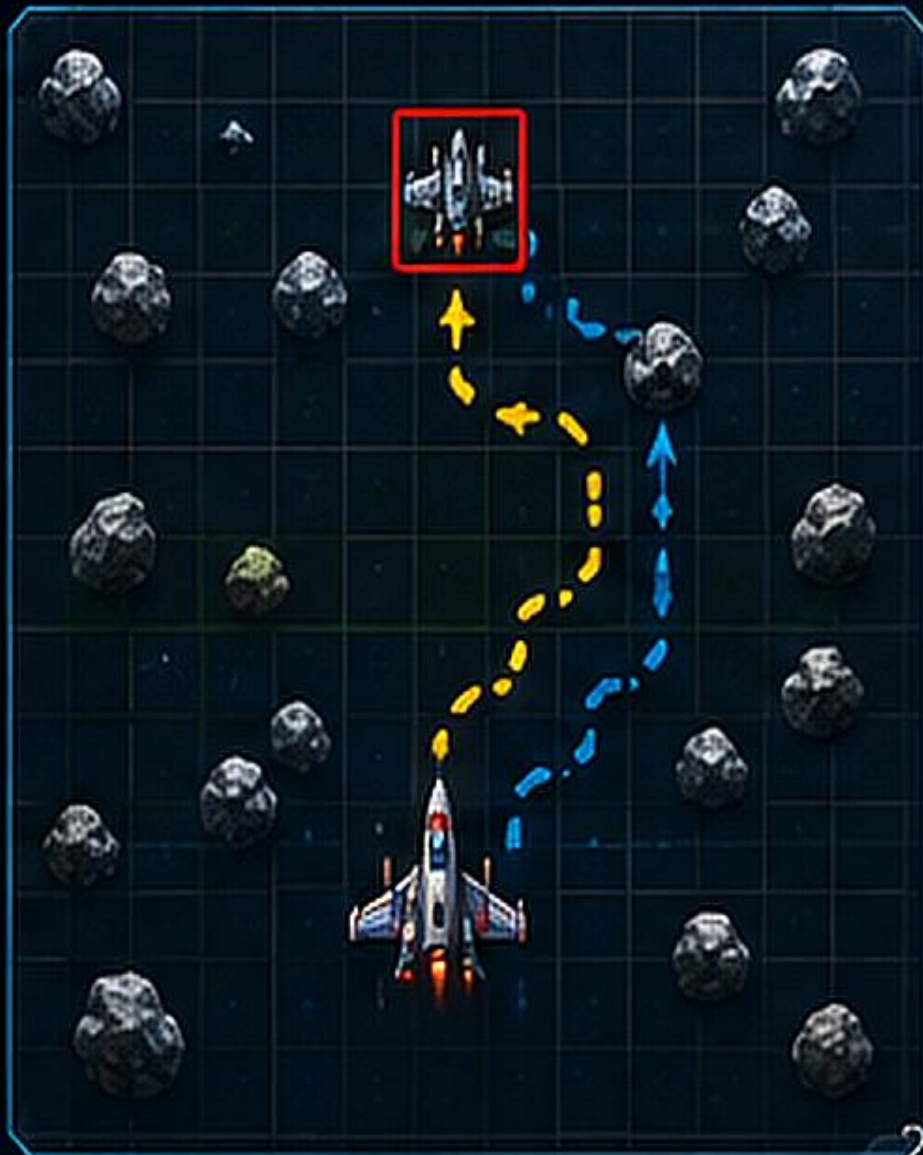


1. CHALLENGE OVERVIEW >>>

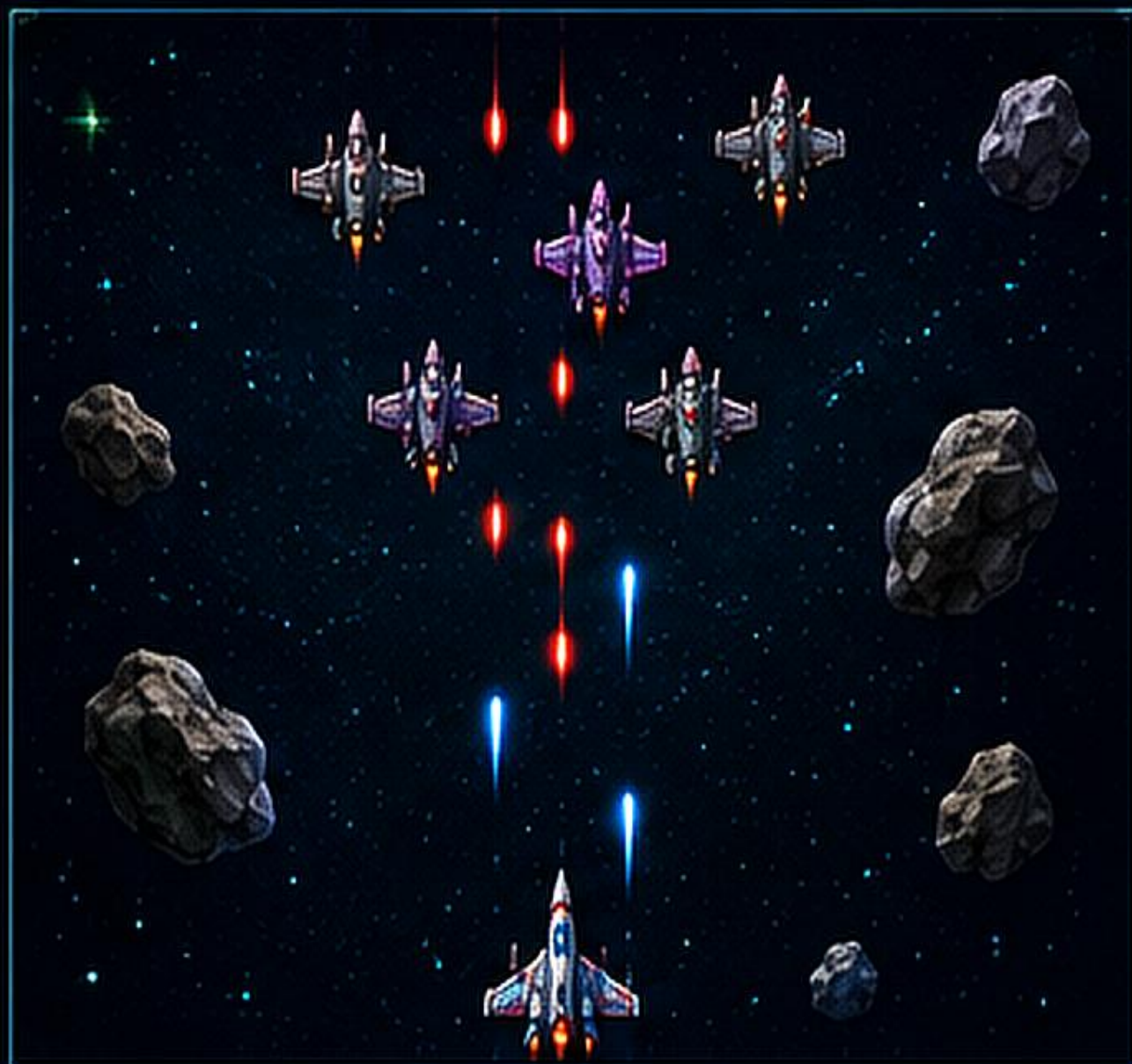
Build a 2D airplane shooting game in Java using Object-Oriented Programming. The enemy aircrafts are not simple straight-line movers — they will intelligently chase the player using **BFS (Breadth-First Search)** pathfinding on a grid map!

KEY FEATURES

- ✓ Player can move and shoot.
- ✓ Enemies chase the player using BFS.
- ✓ Obstacles block the enemy's path.
- ✓ Score system and health system.
- ✓ Game Over and Restart.



2. GAME SCREENSHOT



SCORE

1250

HP



LEVEL

3

CONTROLS



Move

SPACE

Shoot

R

Restart

P

Pause

3. CHALLENGE REQUIREMENTS



Game Window

Create a playable game window using Java Swing.



Player Airplane

Control the player airplane with keyboard (WASD or Arrow Keys). Press SPACE to shoot bullets.



Enemy AI (BFS)

Enemies are not allowed to move in straight lines. They must use BFS to find the shortest path to the player on a grid map.



Obstacles

The map contains obstacles that block movement. Enemies must find another path.



Collision System

Bullets can destroy enemies. Enemies collide with player and reduce HP.



Game System

Score increases when enemies are destroyed. Game Over when HP drops to 0.

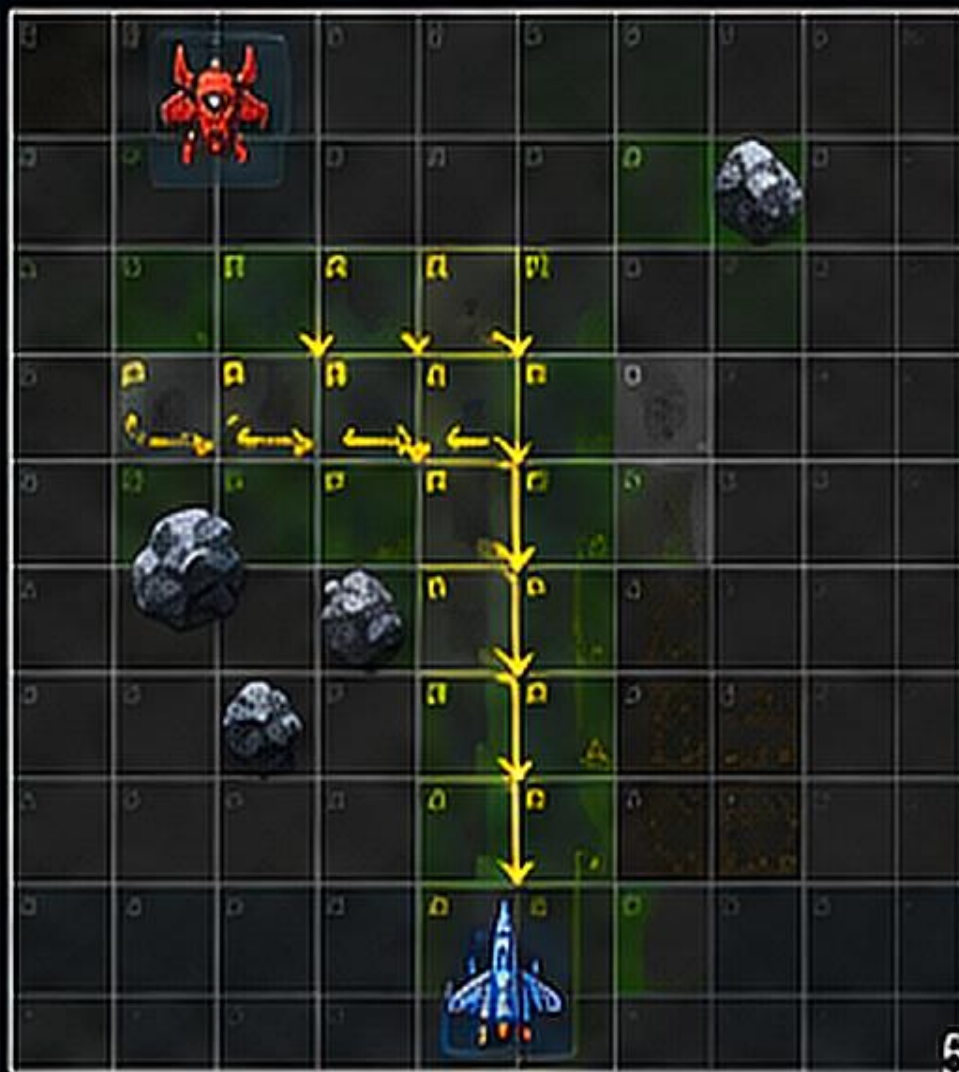
4. BFS PATHFINDING (ENEMY AI) >>

Enemies use BFS to search the shortest path to the player.

1. Treat the map as a grid.
2. Start from the enemy position.
3. Use BFS to visit neighbors (Up, Down, Left, Right).
4. Stop when the player is found.
5. Move one step along the path.



BFS guarantees the shortest path in an unweighted grid!



5. OBJECT-ORIENTED DESIGN

Use OOP to make your code clean, easy to maintain, and scalable.

Main Classes (Example)

GameFrame	// Main window
GamePanel	// Game loop & rendering
Player	// Player airplane
Enemy	// Enemy airplane (BFS AI)
Bullet	// Bullet object
Map	// Grid map & obstacles
Node	// BFS node (x, y)
ScoreBoard	// Score, HP, Level
CollisionManager	// Collision detection



6. DEVELOPMENT ENVIRONMENT



Visual Studio Code
Recommended IDE



Java 17+
Swing (javax.swing)

No external libraries required!

Project Structure (Example)



7. SCORING & RULES



Destroy **enemies**
+100 points each



Survive longer
Bonus score over time



Keep your HP
Lose 1 HP when hit by enemy



Game Over
When HP = 0



Restart
Press **R** to restart the game



8. EVALUATION CRITERIA

Game Functionality 20% ☆☆☆☆☆

BFS Implementation 25% ☆☆☆☆☆

OOP Design 20% ☆☆☆☆☆

Code Quality 15% ☆☆☆☆☆

Game Experience (UI/UX) 20% ☆☆☆☆☆

Submission Requirements

- ✓ Source code (Java)
- ✓ README file (How to run)
- ✓ Screenshot or Video
- ✓ Short description of BFS AI implementation

**CODE YOUR WARPLANE.
OUTSMART THE ENEMY.
WIN THE CHALLENGE!**

